

Given an integer array **arr[]**, find the **subarray** (containing at least one element) which has the **maximum possible sum**, and return that sum.

Note: A **subarray** is a continuous part of an array.

Examples:

Input: `arr[] = [2, 3, -8, 7, -1, 2, 3]`

Output: 11

Explanation: The subarray [7, -1, 2, 3] has the largest sum 11.

Input: `arr[] = [-2, -4]`

Output: -2

Explanation: The subarray [-2] has the largest sum -2.

Input: `arr[] = [5, 4, 1, 7, 8]`

Output: 25

Explanation: The subarray [5, 4, 1, 7, 8] has the largest sum 25.

Given an array, find a subarray with the max

Kadane's Algorithm

Scenario 1	subarray sum (+ive)	current (+ive) → Extend
Scenario 2	subarray sum (-ive)	current (+ive) → New
Scenario 3	subarray sum (+ive)	current (-ive) → Extend
Scenario 4	subarray sum (-ive)	current (-ive) → New